

Amazon CloudWatch Incident Investigation

Project Evidence and Implementation Screenshots

This document compiles the key AWS Console screenshots used to demonstrate EC2 logging, CloudWatch Logs, Logs Insights investigation, metric filtering, alarm-based detection, SNS email notification, and dashboard monitoring.

Project flow: EC2 → CloudWatch Agent → CloudWatch Logs → Logs Insights → Metric Filter → Alarm → SNS → Dashboard

Screenshot 1 - EC2 Instance Running

The screenshot displays the AWS Management Console interface for the EC2 service. The left-hand navigation pane shows the 'EC2' section selected, with sub-options like 'Dashboard', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'UltraServers', and 'Capacity Manager'. The main content area is titled 'Instances (1) Info' and shows a single instance in a 'Running' state. The instance details table lists the following information:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name
CloudWatch-I...	i-0b8a816ae701eca84	Running	t3.micro	3/3 checks passed	eu-north-1b	ec2-16-171-150-114.eu...	16.171.150.114	-	-	disabled	launch-wizard-2

Below the table, the 'All instances' section provides a summary of the instance's health and status. It includes three panels:

- Instance alarm states:** Shows 'In alarm' as 'None' with a link to 'View in CloudWatch'.
- Service health:** Shows 'EC2 service status' as 'Operating normally' with a link to 'View Service Health Dashboard'.
- Health events affecting EC2 instances:** Shows 'Open events' and 'Upcoming events' as 'None' with links to 'View AWS Health Dashboard'.

The bottom of the console shows the footer with copyright information and links to 'Privacy', 'Terms', and 'Cookie preferences'.

Shows the Amazon EC2 instance used for the incident-investigation project running successfully.

Source file: 01-EC2-running.png

Screenshot 2 - IAM Role and CloudWatch Policy

The screenshot displays the AWS IAM console interface. The left-hand navigation pane shows the 'Identity and Access Management (IAM)' section, with 'Roles' selected. The main content area shows the details for the 'CloudWatchAgentEC2Role'. The role's summary includes its creation date (October 02, 2026, 16:45 UTC+05:30), last activity (24 minutes ago), ARN (arn:aws:iam::306255359410:role/CloudWatchAgentEC2Role), and instance profile ARN (arn:aws:iam::306255359410:instance-profile/CloudWatchAgentEC2Role). The 'Permissions' tab is active, showing one attached policy: 'CloudWatchAgentServerPolicy' (AWS managed). The console also features a 'Generate policy based on CloudTrail events' section at the bottom.

CloudWatchAgentEC2Role [Info](#) [Delete](#)

Allows EC2 instances to call AWS services on your behalf.

Summary [Edit](#)

Creation date
October 02, 2026, 16:45 (UTC+05:30)

ARN
[arn:aws:iam::306255359410:role/CloudWatchAgentEC2Role](#)

Instance profile ARN
[arn:aws:iam::306255359410:instance-profile/CloudWatchAgentEC2Role](#)

Last activity
[24 minutes ago](#)

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (1) [Info](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 20 managed policies.

Filter by Type
All types

☐	Policy name ↗	Type	Attached entities
☐	CloudWatchAgentServerPolicy	AWS managed	1

► **Permissions boundary** (not set)

▼ **Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#) [↗](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

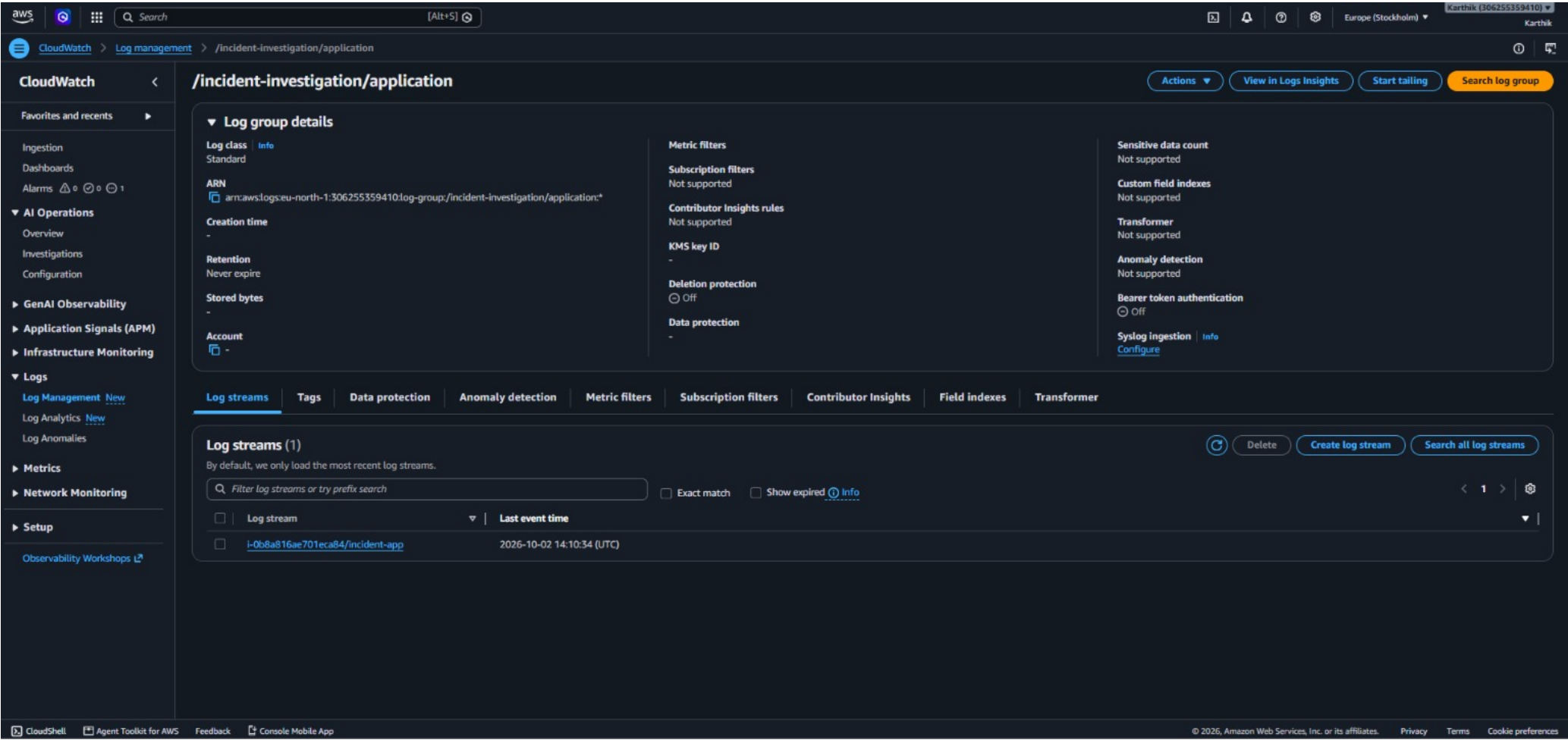
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Shows the CloudWatchAgentEC2Role and the attached CloudWatchAgentServerPolicy permission.

Source file: 02-IAM-role.png

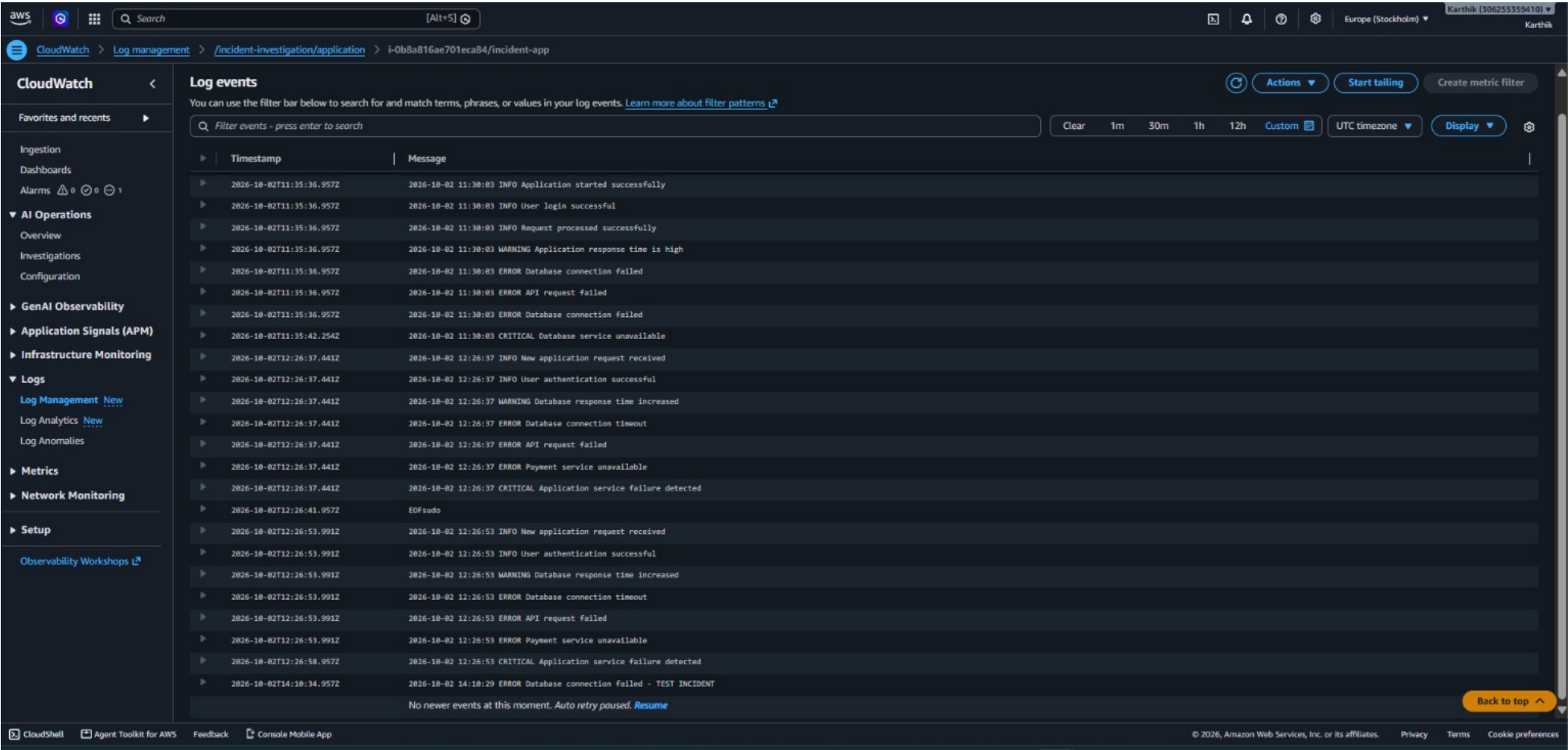
Screenshot 3 - CloudWatch Log Group



Shows the /incident-investigation/application CloudWatch log group and its log stream.

Source file: 03-CloudWatch-log-group.png

Screenshot 4 - CloudWatch Log Events



Shows application log events including INFO, WARNING, ERROR, and CRITICAL entries.

Source file: 04-CloudWatch-logs.png

Screenshot 5 - CloudWatch Logs Insights

The screenshot displays the AWS CloudWatch Logs Insights interface. At the top, a notification banner states: "You've been redirected to the new Log Analytics experience. Log Analytics combines Logs Insights, Live Tail, and Contributor Insights in a unified experience. Clicking OK sets Log Analytics as your default going forward. Opting out returns you to the prior experience. Manage this preference anytime from the top-right preferences menu." Below this, there's a search bar with "Log Insights" and an "Ask Amazon Q" button. A "Query scope" section includes a filter "Log group" set to "/incident-investigation/application" and a time range of "Last 604800 seconds". The query itself is:

```
1 SOURCE "/arn:aws:logs:eu-north-1:306255359410:log-group:/incident-investigation/application" START--604800s END--0s |
2 fields @timestamp, @message
3 | sort @timestamp asc
4 | limit 20
```

 The interface shows "Discovered fields (16)" and "Lookup tables". A status bar indicates "Complete: Showing 17 of 17 matched. Query executed for 1 log group." and "17 records (1.4 KB) scanned in 1.3s @ 13 records/s (1.1 KB/s)". A table of log records is displayed with columns "@timestamp" and "@message". The records show various events, including database service unavailability, application requests, connection timeouts, authentication success, response time increases, API request failures, application service failures, and payment service unavailability. The bottom of the screen shows navigation links for CloudShell, Agent Toolkit for AWS, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services.

aws Europe (Stockholm) Karthik (306255359410) Karthik

ⓘ You've been redirected to the new Log Analytics experience. Log Analytics combines Logs Insights, Live Tail, and Contributor Insights in a unified experience. Clicking OK sets Log Analytics as your default going forward. Opting out returns you to the prior experience. Manage this preference anytime from the top-right preferences menu.

▼ Ask AI to write a query... Saved queries Query History

Query scope ⓘ

5m 30m 1h 3h 12h Last 604800 seconds UTC timezone ▼

Log group ▼ = /incident-investigation/application + Add filter

- Collapse filters Reset to default

```
1 SOURCE "/arn:aws:logs:eu-north-1:306255359410:log-group:/incident-investigation/application" START--604800s END--0s |
2 fields @timestamp, @message
3 | sort @timestamp asc
4 | limit 20
```

Add stats to visualize Analyze patterns Estimated bytes scanned: 1.4 KB Live tail Run

Discovered fields (16) ⓘ Lookup tables ⓘ

Complete: Showing 17 of 17 matched. Query executed for 1 log group. 17 records (1.4 KB) scanned in 1.3s @ 13 records/s (1.1 KB/s)

Filter: Highlight Table ▼ Actions ▼

@timestamp	@message
> 2026-10-02T11:35:42.254Z	2026-10-02 11:30:03 CRITICAL Database service unavailable
> 2026-10-02T12:25:37.441Z	2026-10-02 12:26:37 INFO New application request received
> 2026-10-02T12:26:37.441Z	2026-10-02 12:26:37 ERROR Database connection timeout
> 2026-10-02T12:26:37.441Z	2026-10-02 12:26:37 INFO User authentication successful
> 2026-10-02T12:26:37.441Z	2026-10-02 12:26:37 WARNING Database response time increased
> 2026-10-02T12:26:37.441Z	2026-10-02 12:26:37 ERROR API request failed
> 2026-10-02T12:26:37.441Z	2026-10-02 12:26:37 CRITICAL Application service failure detected
> 2026-10-02T12:26:37.441Z	2026-10-02 12:26:37 ERROR Payment service unavailable
> 2026-10-02T12:26:41.957Z	EOF sudo
> 2026-10-02T12:26:53.991Z	2026-10-02 12:26:53 INFO New application request received
> 2026-10-02T12:26:53.991Z	2026-10-02 12:26:53 WARNING Database response time increased
> 2026-10-02T12:26:53.991Z	2026-10-02 12:26:53 ERROR API request failed
> 2026-10-02T12:26:53.991Z	2026-10-02 12:26:53 ERROR Payment service unavailable
> 2026-10-02T12:26:53.991Z	2026-10-02 12:26:53 INFO User authentication successful
> 2026-10-02T12:26:53.991Z	2026-10-02 12:26:53 ERROR Database connection timeout
> 2026-10-02T12:26:58.957Z	2026-10-02 12:26:53 CRITICAL Application service failure detected
> 2026-10-02T14:10:34.957Z	2026-10-02 14:10:29 ERROR Database connection failed - TEST INCIDENT

17 logs ⓘ

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Shows Logs Insights querying the application log group and returning log records.

Source file: 05-Logs-Insights.png

Screenshot 6 - ERROR and CRITICAL Investigation

The screenshot displays the AWS Logs Insights console interface. At the top, there's a navigation bar with the AWS logo, a search bar containing 'Logs Insights', and a button to 'Ask Amazon Q'. Below this, a message states: 'You've been redirected to the new Log Analytics experience. Log Analytics combines Logs Insights, Live Tail, and Contributor Insights in a unified experience. Clicking OK sets Log Analytics as your default going forward. Opting out returns you to the prior experience. Manage this preference anytime from the top-right preferences menu.' There are 'OK' and 'Opt out (Logs Insights)' buttons.

The main area features a 'Query scope' section with a search bar 'Search to add a filter...' and a dropdown menu for 'Log group' set to '/incident-investigation/application'. Below this, a query is entered in a text area:

```
1 SOURCE "arn:aws:logs:eu-north-1:306255359410:log-group:/incident-investigation/application" START=-604800s END=0s |
2 fields @timestamp, @message
3 | filter @message like /ERROR|CRITICAL/
4 | sort @timestamp desc
```

Buttons for 'Add stats to visualize', 'Analyze patterns', 'Live tail', and 'Run' are visible. The 'Run' button is highlighted in yellow. Below the query, a bar chart shows the distribution of log events over time, with a peak on October 2nd. The x-axis is labeled with dates from Sep 26 to Oct 2. The y-axis represents the number of records, ranging from 0 to 8. A legend indicates '17 records (1.4 KB) scanned in 1.2s @ 14 records/s (1.1 KB/s)'.

Below the chart, a table displays the results of the query. The table has columns for '@timestamp' and '@message'. The results show several log entries, including 'ERROR Database connection failed - TEST INCIDENT', 'CRITICAL Application service failure detected', 'ERROR Database connection timeout', 'ERROR Payment service unavailable', and 'ERROR API request failed'. The table is filtered by '@timestamp' and '@message'.

At the bottom of the console, there's a footer with links to 'CloudShell', 'Agent Toolkit for AWS', 'Feedback', and 'Console Mobile App'. The footer also includes copyright information: '© 2026, Amazon Web Services, Inc. or its affiliates.' and links to 'Privacy', 'Terms', and 'Cookie preferences'.

Shows the Logs Insights query filtering ERROR and CRITICAL incidents.

Source file: 06-Error-query.png

Screenshot 7 - Metric Filter

The screenshot displays the AWS CloudWatch console interface. The left-hand navigation pane shows the 'CloudWatch' logo and a list of services including 'Log management', 'AI Operations', 'GenAI Observability', 'Application Signals (APM)', 'Infrastructure Monitoring', 'Logs', 'Metrics', 'Network Monitoring', and 'Setup'. The 'Logs' section is expanded, showing 'Log Management' as the active service. The main content area is titled 'Incident investigation/application' and features a top navigation bar with tabs for 'Log streams', 'Tags', 'Data protection', 'Anomaly detection', 'Metric filters' (which is selected), 'Subscription filters', 'Contributor Insights', 'Field indexes', and 'Transformer'. Below the tabs, the 'Metric filters (1)' section is visible, containing a search bar and a list of filters. The 'ApplicationErrorFilter' is the only filter listed, with a checkbox to its right. The filter details are as follows:

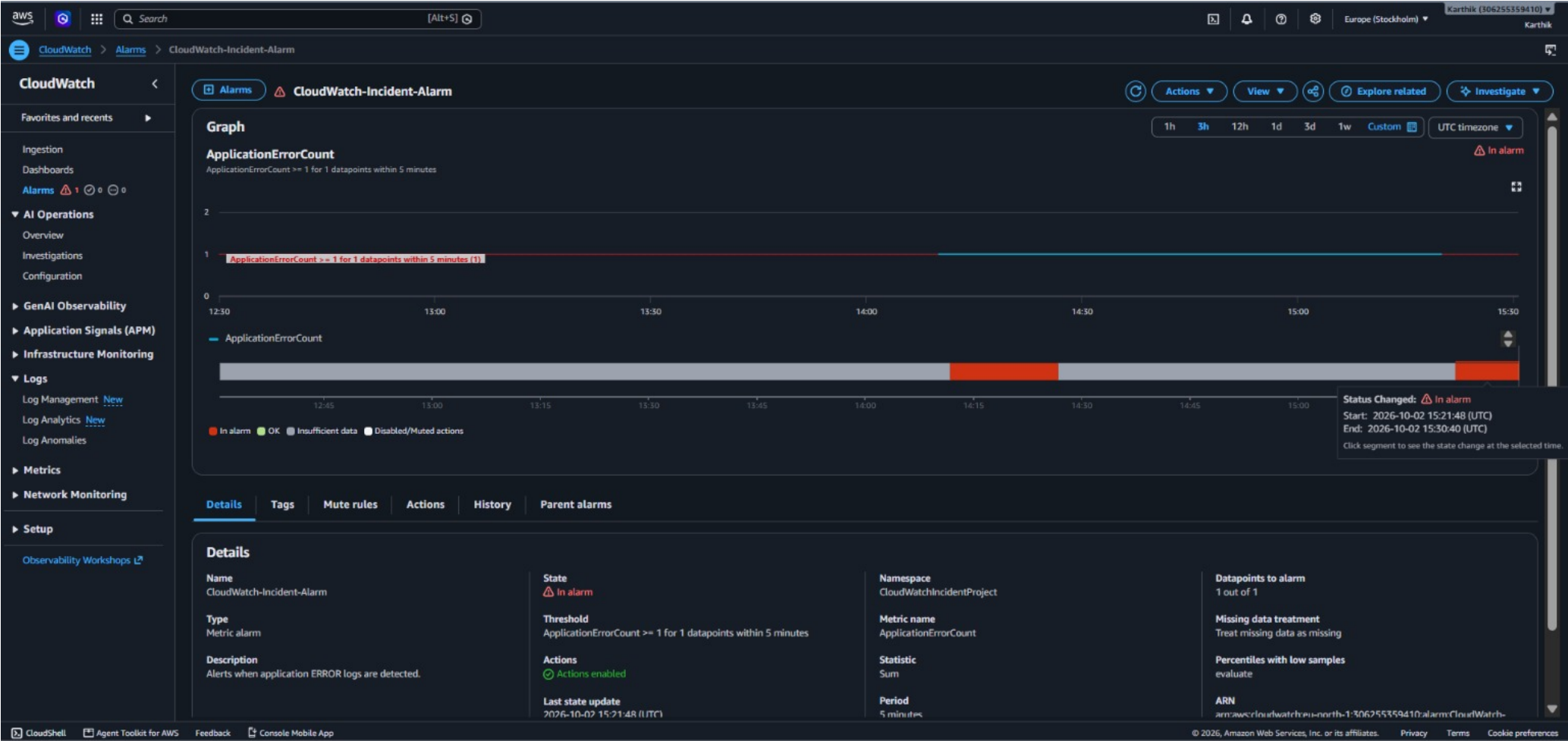
- Filter pattern:** ERROR
- Field selection criteria:** -
- Metric:** CloudWatchIncidentProject / ApplicationErrorCount
- Metric value:** 1
- Default value:** 0
- Applied on transformed logs:** -
- Unit:** None
- Emit system field dimensions:** -
- Dimensions:** -
- Alarms:** CloudWatch-Incident-Alarm

The bottom of the console shows the footer with links to 'CloudShell', 'Agent Toolkit for AWS', 'Feedback', and 'Console Mobile App', along with the copyright notice '© 2026, Amazon Web Services, Inc. or its affiliates.' and links to 'Privacy', 'Terms', and 'Cookie preferences'.

Shows ApplicationErrorFilter with the ERROR pattern and the ApplicationErrorCount metric.

Source file: 07-Metric-filter.png

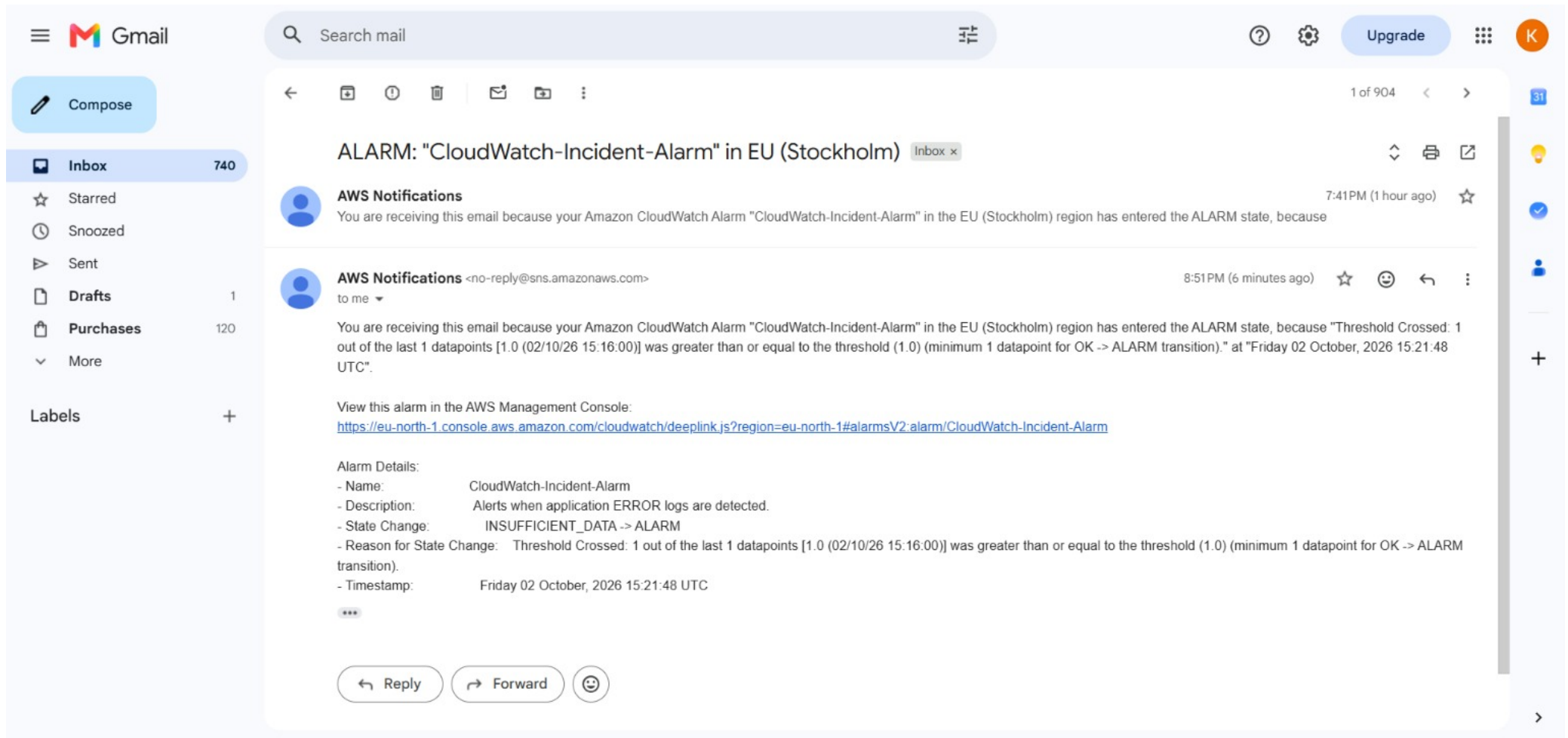
Screenshot 8 - CloudWatch Alarm in ALARM State



Shows CloudWatch-Incident-Alarm in the ALARM state with the ApplicationErrorCount threshold.

Source file: 10 cloud watch incident alarm.jpeg

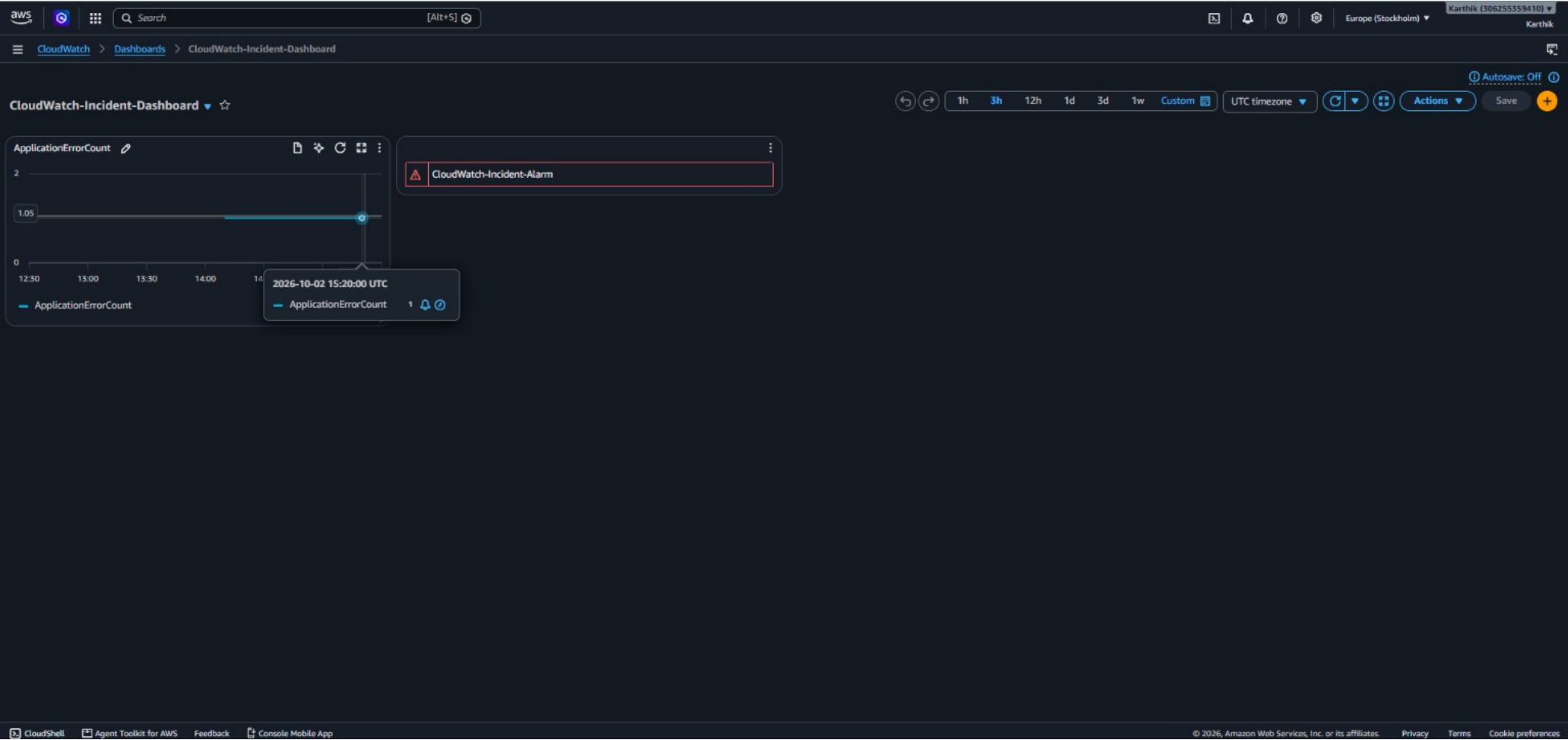
Screenshot 9 - SNS Email Notification



Shows the AWS SNS/CloudWatch alarm notification received by email and the threshold-crossed details.

Source file: 09-SNS-email.png

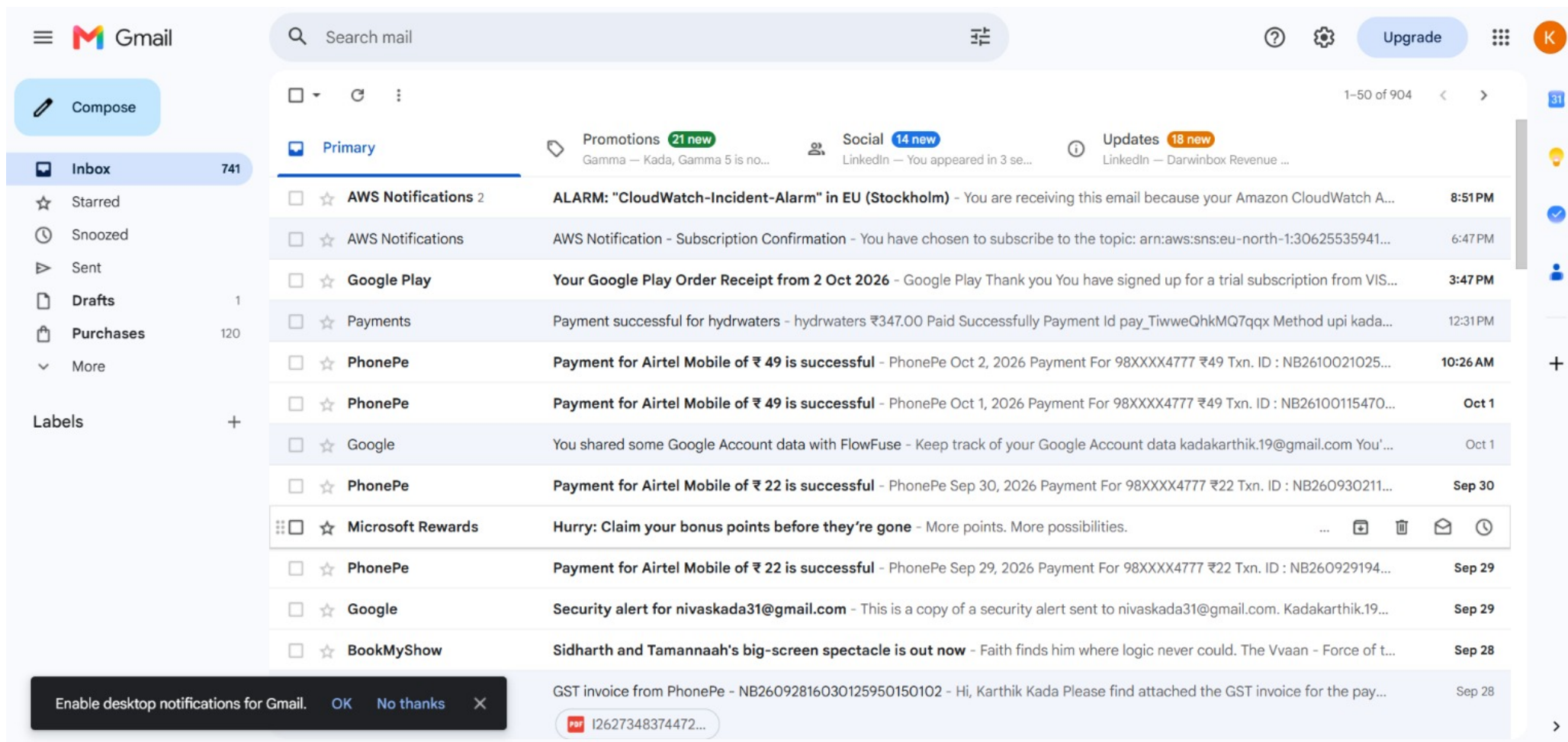
Screenshot 10 - CloudWatch Dashboard



Shows the CloudWatch-Incident-Dashboard with the ApplicationErrorCount graph and alarm status.

Source file: 11-Dashboard.png

Additional Evidence - SNS Alarm Notification in Gmail Inbox



Additional screenshot showing the CloudWatch alarm notification visible in the Gmail inbox.

Source file: 08-Alarm-ALARM.png